

Model Discordance as a Clinical Complexity Marker in Type 2 Diabetes Risk Prediction

Group 3: Mohammad Shaan, Fatin Khan, Sai Han Htun,
Bartłomiej Mazuruk

SCHOOL OF BIOSAFETY
SUN YAT-SEN UNIVERSITY
INTERNATIONAL SUMMER SCHOOL ON HEALTH MEDICAL BIG DATA AND INTELLIGENT
STATISTICS

AUGUST 2026

Introduction

Clinical Challenge

- Type 2 Diabetes risk prediction has become central to preventive medicine.
- Numerous statistical and machine learning models achieve comparable overall performance.
- However, different models often assign different risks to the same patient.

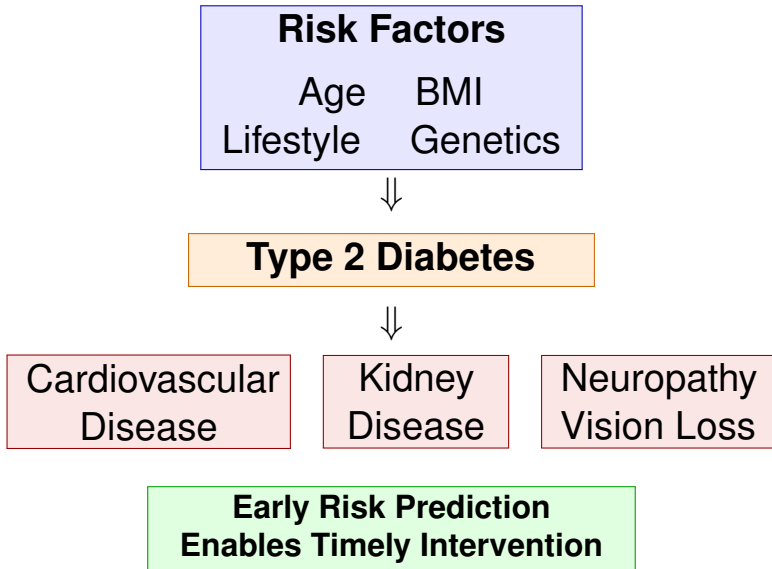
Research Question

Can disagreement between prediction models serve as a marker of clinical complexity?

Type 2 Diabetes: Beyond High Blood Sugar

- **One of the world's most prevalent chronic metabolic diseases**, affecting over **500 million** adults globally.
- Associated with serious long-term complications including **cardiovascular disease, chronic kidney disease, neuropathy, and vision loss**.
- **Early identification of high-risk individuals** enables timely lifestyle intervention and targeted clinical management.
- Consequently, **accurate risk prediction** has become a cornerstone of preventive medicine.

Why Early Risk Prediction Matters



Study Design

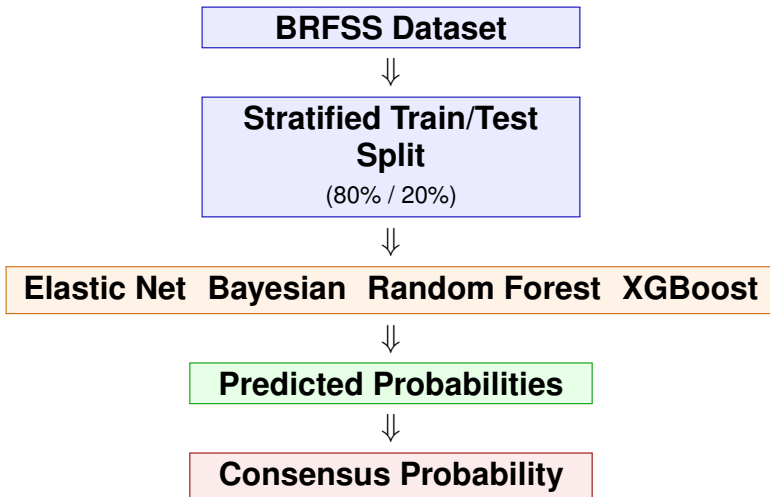
Dataset

- BRFSS 2015 (CDC)
- National cross-sectional survey of US adults
- Outcome: Self-reported Type 2 Diabetes (Yes/No)
- Stratified train-test split (80% / 20%)

Preprocessing

- Clinical feature engineering
- One-hot encoding of categorical predictors
- Preprocessing estimated on training data only
- Applied identically to test data (no information leakage)

Prediction Pipeline



Quantifying Model Discordance

Consensus Prediction

$$\bar{P} = \frac{P_{\text{ElasticNet}} + P_{\text{Bayesian}} + P_{\text{RandomForest}} + P_{\text{XGBoost}}}{4}$$

Discordance Score

$$D_i = \sqrt{\frac{1}{K-1} \sum_{k=1}^K (p_{ik} - \bar{P}_i)^2}$$

where

$$K = 4 \text{ models.}$$

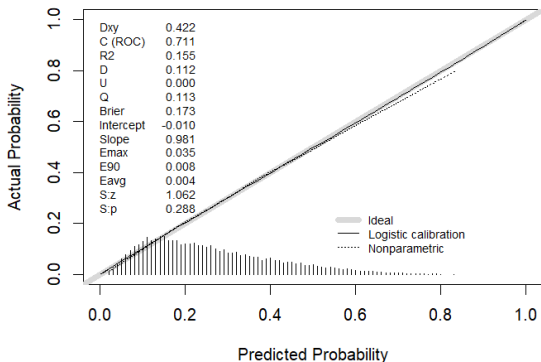
- Low discordance → models agree
- High discordance → substantial disagreement
- Discordance categorized into Low, Medium and High tertiles

Predictive Performance Across Statistical Paradigms

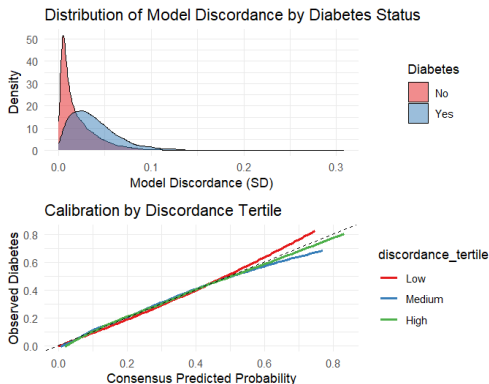
Model Comparison

Model	AUC
Elastic Net	0.711
Bayesian Logistic	0.782
Random Forest	0.855
XGBoost	0.828
Consensus	0.828

- All models achieved acceptable predictive performance.
- Ensemble methods showed superior discrimination.
- Consensus prediction was used for downstream analyses.



Model Discordance Reflects Clinical Heterogeneity



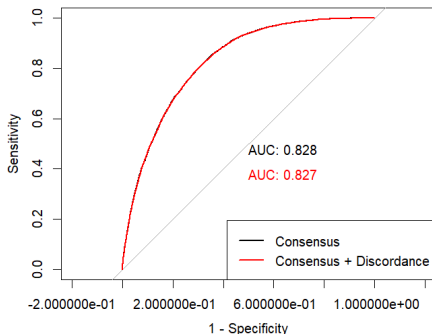
Key Findings

- Greater discordance observed among diabetic participants.
- Participants were stratified into Low, Medium and High discordance tertiles.
- Calibration patterns differed across discordance groups.

Basic Idea

Model disagreement is not random and appears concentrated among clinically higher-risk individuals.

Does Discordance Improve Prediction?



Likelihood Ratio Test

$$\chi^2 = 13.75$$

$$p = 2.09 \times 10^{-4}$$

Discrimination

Consensus

AUC = 0.828

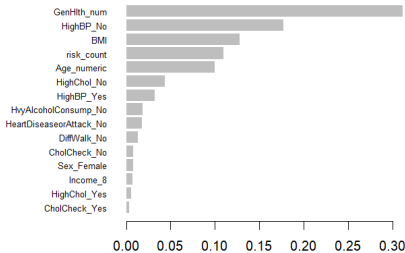
Consensus + Discordance

AUC = 0.827

Interpretation

Discordance improves statistical model fit but contributes minimal additional discriminatory performance.

Clinical Drivers of Model Discordance



Most Influential Variables

- 1 General Health**
- 2 Blood Pressure**
- 3 BMI**
- 4 Clinical Risk Burden**
- 5 Age**

Discordance was no longer independently associated with diabetes after adjustment for these established clinical factors.

Clinical Interpretation

Model disagreement primarily reflects underlying clinical complexity already captured by conventional risk factors.

Conclusions

Key Findings

- Consensus prediction demonstrated good discrimination and calibration.
- Model discordance was greater among clinically higher-risk individuals.
- Discordance improved statistical model fit but provided limited additional discrimination.

Clinical Implications

- Model disagreement reflects underlying clinical heterogeneity.
- Much of this information overlaps with established clinical risk factors.
- Discordance may serve as a complementary indicator for identifying clinically complex patients.

**Model disagreement may provide complementary clinical insight,
even when predictive accuracy remains unchanged.**

Thank You